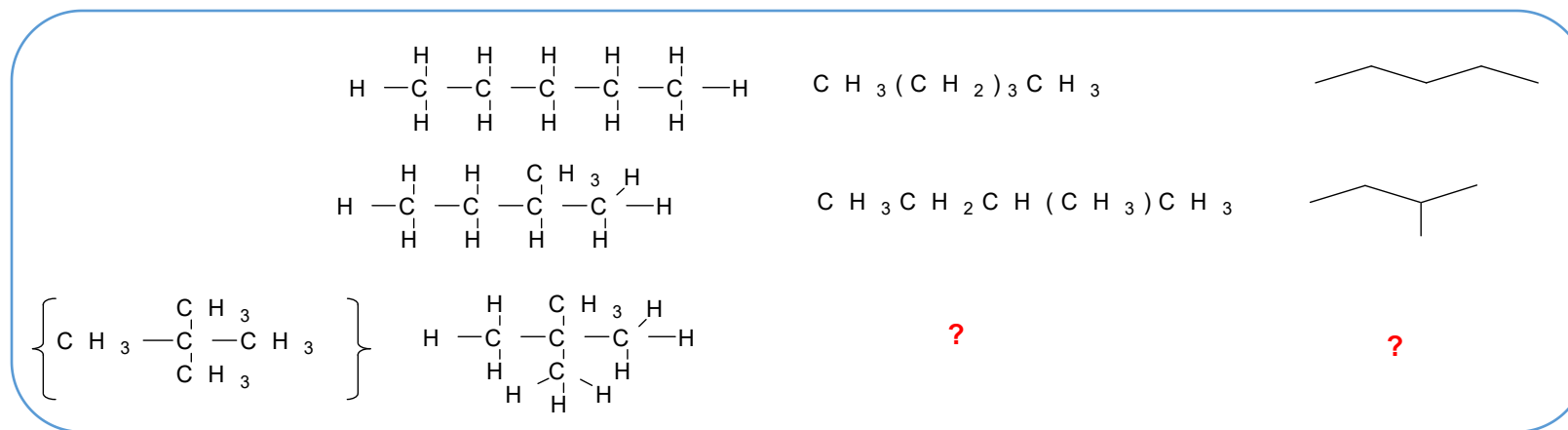
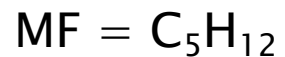


Notes 4

Structural Isomerism

Structural isomerism is the type of isomerism where the atoms and the functional groups are joined together in different ways



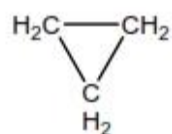
Give the IUPAC names of the above alkanes

Assignment

- How many structural isomers does C_6H_{14} have?
- Draw all the structural isomers for C_6H_{14}
- Give the IUPAC name for each of the structural isomers of C_6H_{14}

Monocyclic Alkanes

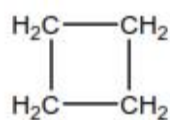
The prefix “cyclo” is used before the name of the cyclic parent alkane. Some examples are:



or



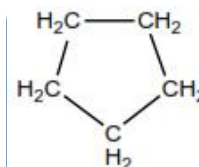
cyclopropane



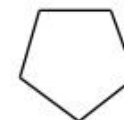
or



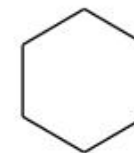
cyclobutane



or

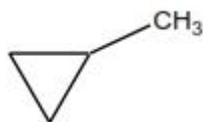


cyclopentane

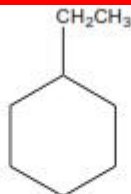


cyclohexane

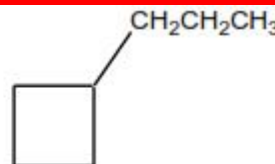
When the cyclic compound has one substituent:



methylcyclopropane

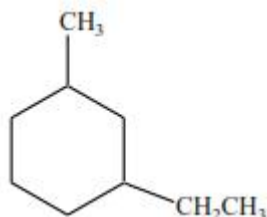


ethylcyclohexane

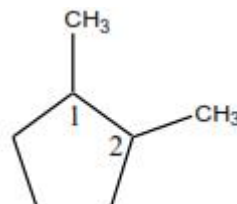


n-propylcyclobutane

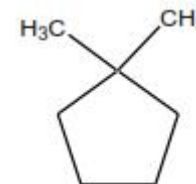
When two substituents are present:



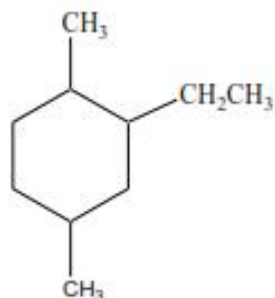
1-ethyl-3-methylcyclohexane
not 1-ethyl-5-methylcyclohexane



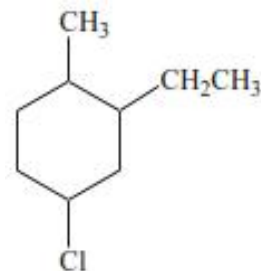
1,2-dimethylcyclopentane
not 1,5-dimethylcyclopentane



1,1-dimethylcyclopentane

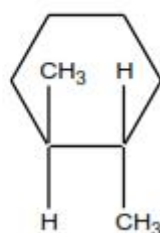
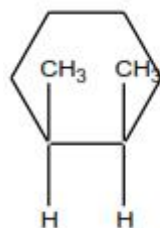


2-ethyl-1,4-dimethylcyclohexane

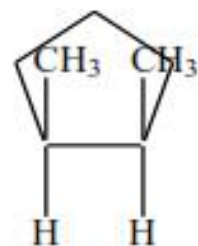


4-chloro-2-ethyl-1-methylcyclohexane

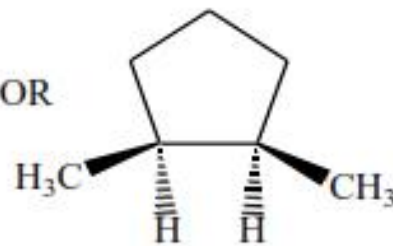
Two ways of arranging the two substituents on the ring are possible -
Indicating the stereochemistry of the substituents (*cis* and *trans*).



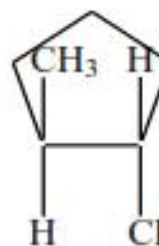
The two are called **stereoisomers**. They have different physical properties.



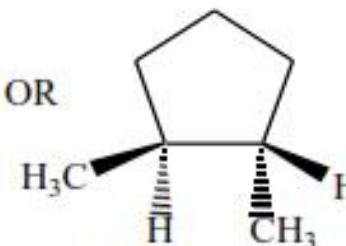
OR



cis-1,2-dimethylcyclopentane
bp 99°C
mp -62°



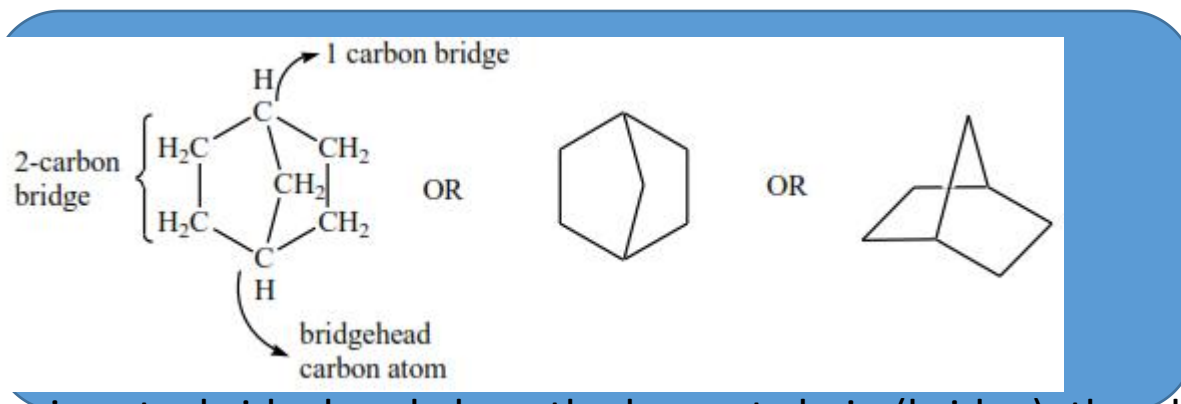
OR



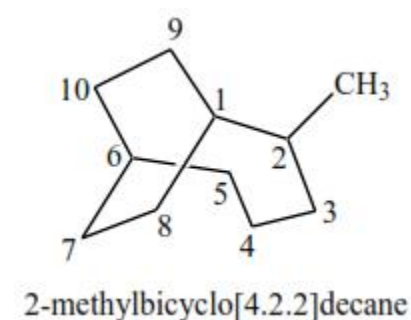
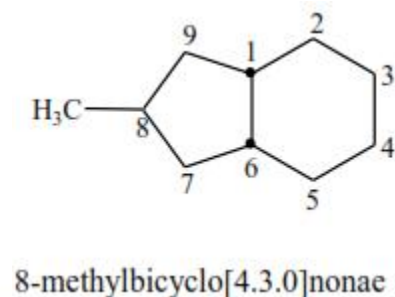
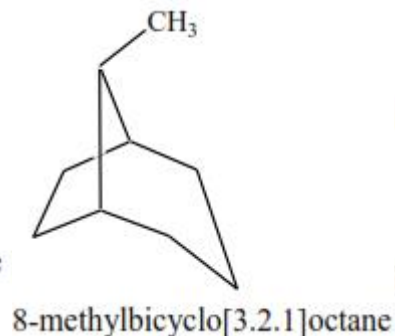
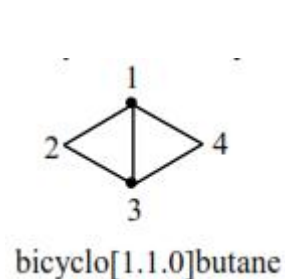
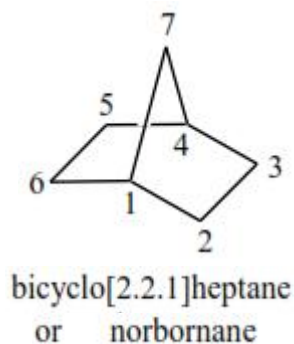
trans-1,2-dimethylcyclopentane
bp 92°C
mp -120°C

Bridged Bicycloalkanes

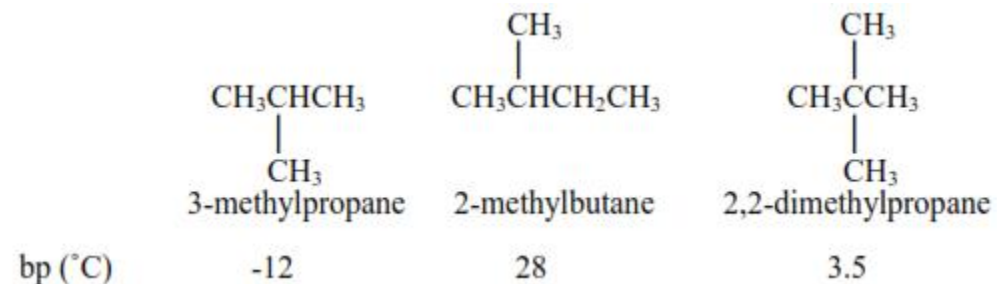
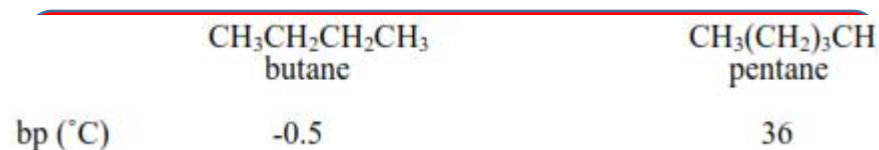
- Bridged bicyclic alkanes have two rings that share two carbon atoms. For example:



- Numbering** begins at a bridgehead along the longest chain (bridge), then back to the first bridgehead by the second longest chain.



PHYSICAL PROPERTIES



Task

- PHYSICAL AND CHEMICAL PROPERTIES OF ALKANES
- <https://www.youtube.com/watch?v=0dgOHSgHKZQ> by DeltaStep
- i) <https://www.youtube.com/watch?v=gW8RI4P3-Rg> by Kamran Maqsood
- ii) <https://www.youtube.com/watch?v=CrGzeISasgk> by myhometuition

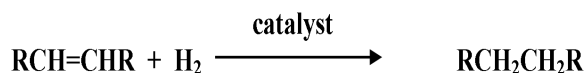
Preparation of alkanes and cycloalkanes

ALKANES

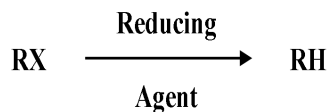
Wurtz reaction



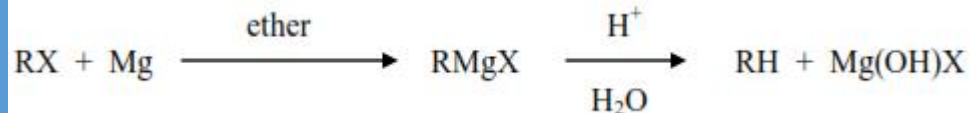
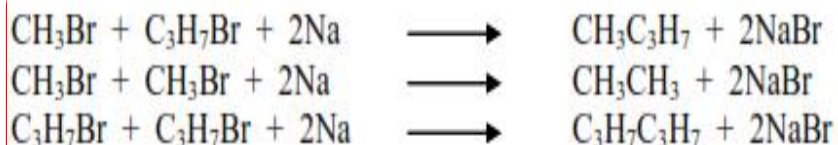
Hydrogenation of unsaturated hydrocarbons



Reduction of alkyl halides



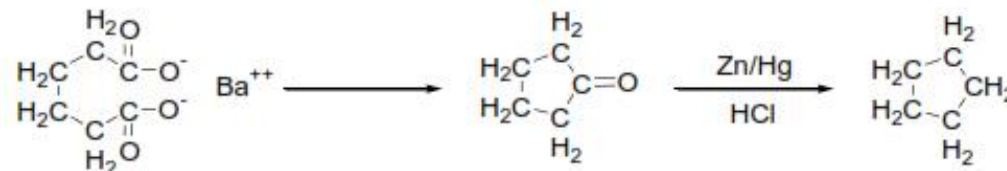
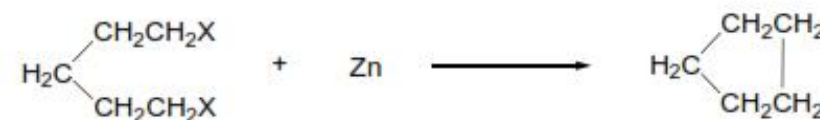
Wurtz Reaction



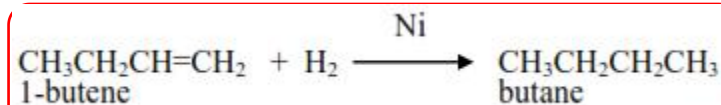
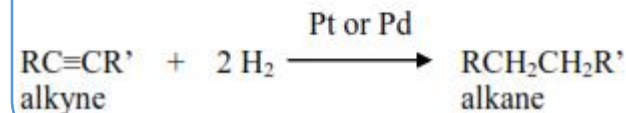
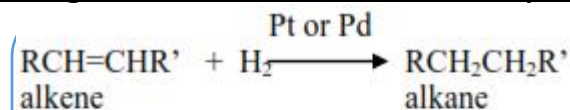
Decarboxylation of carboxylic acids



CYCLOALKANES

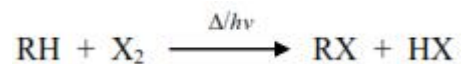


Hydrogenation of Unsaturated Hydrocarbons

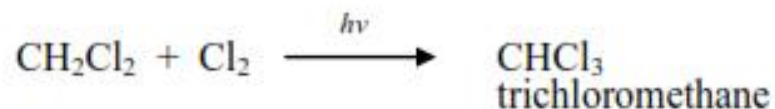
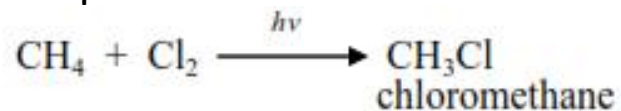


Reactions of alkanes

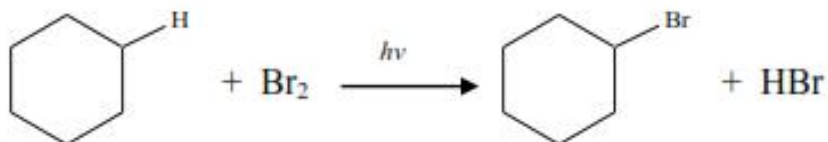
1. Halogenation



Multiple substitutions

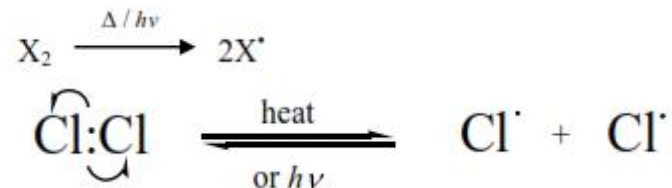


For cycloalkanes

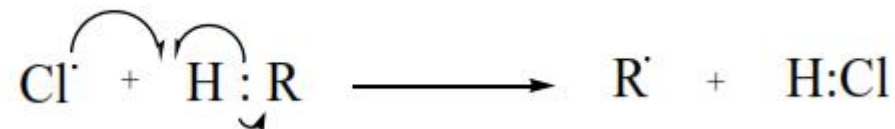


Mechanism of halogenation

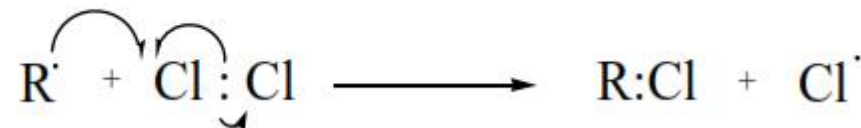
Step (i)



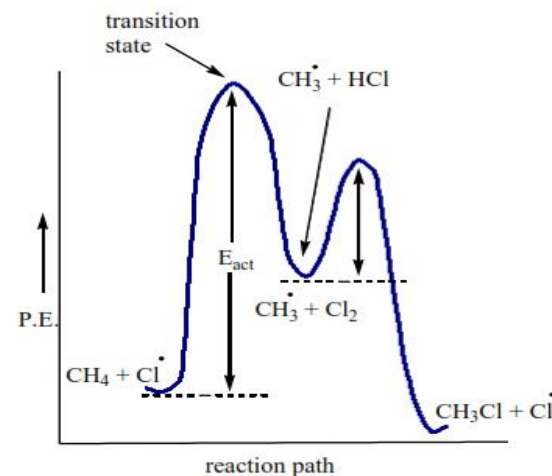
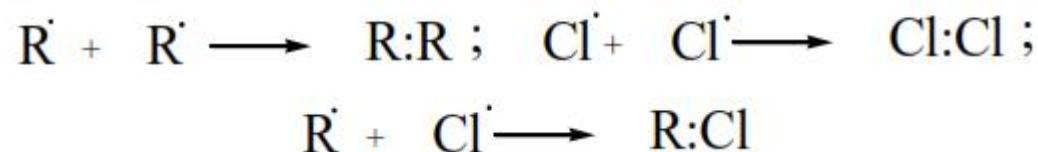
Step (ii)



Step (iii)



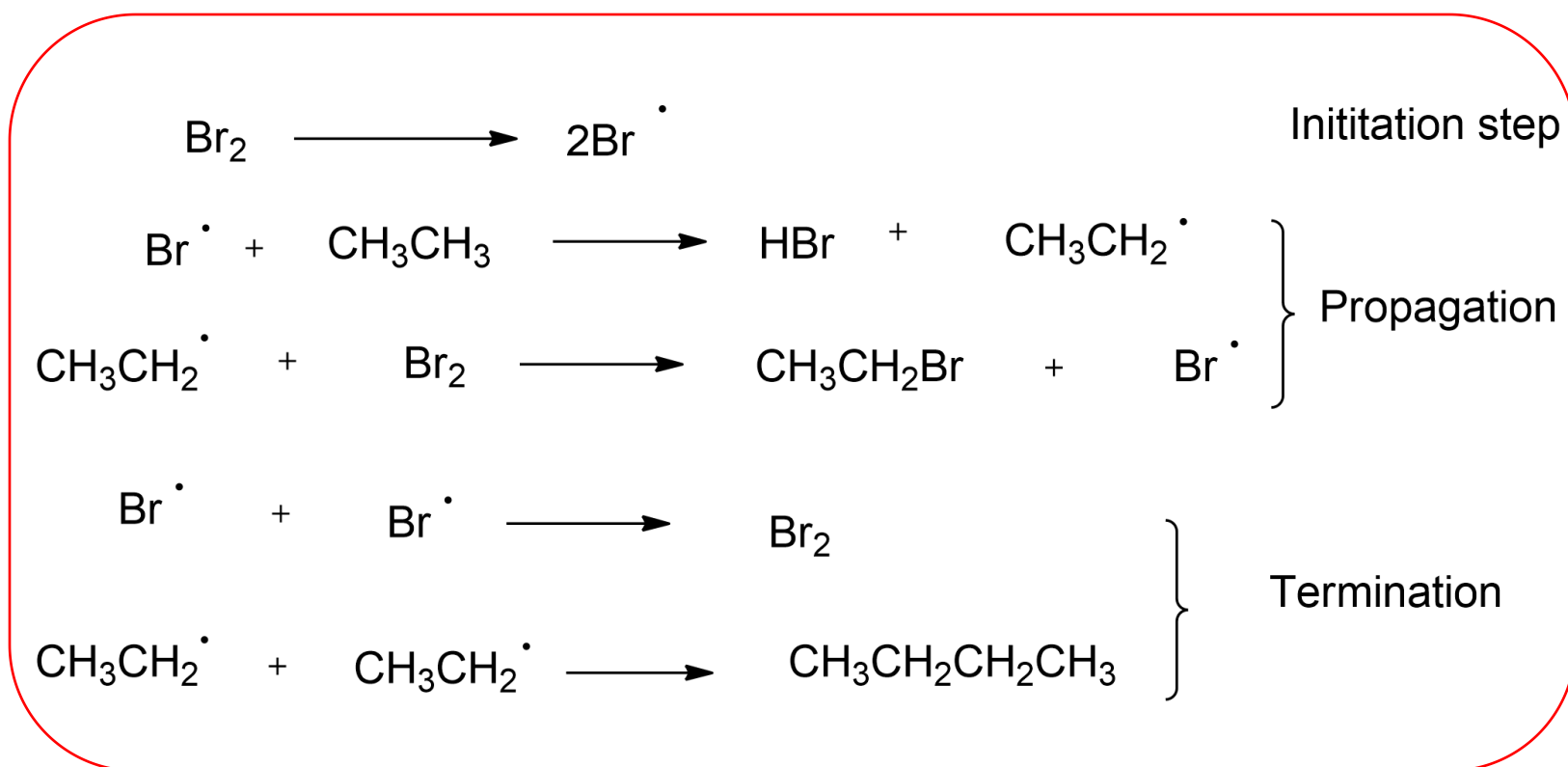
Step (iv)



Energy profile diagram for the reaction of $\text{Cl}^\bullet$ and CH_4 .

- Ethane can be brominated to bromoethane using molecular bromine in the presence of light.

Reaction Mechanism



Recap

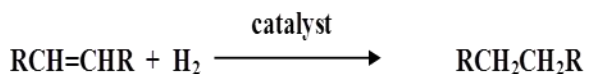
- Nomenclature of Bicycloalkanes – Bridgehead carbon atoms and Bridge Carbon atoms
- Preparation & reactions of Alkanes

ALKANES

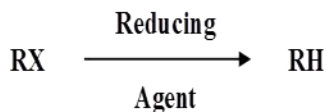
Wurtz reaction



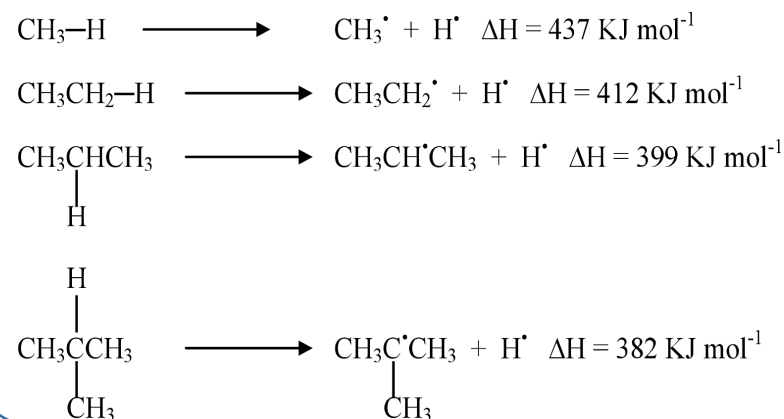
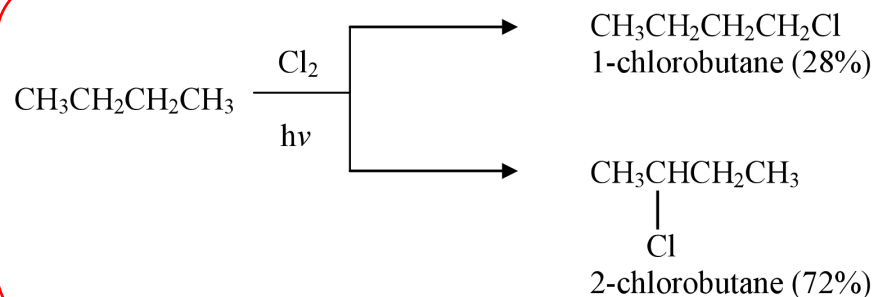
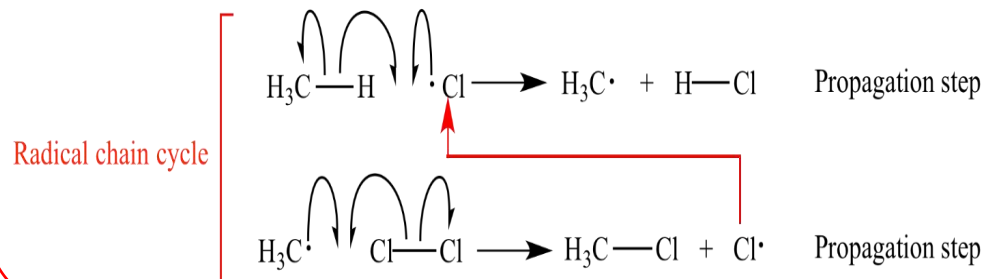
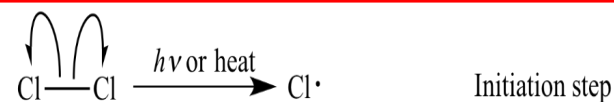
Hydrogenation of unsaturated hydrocarbons



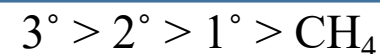
Reduction of alkyl halides



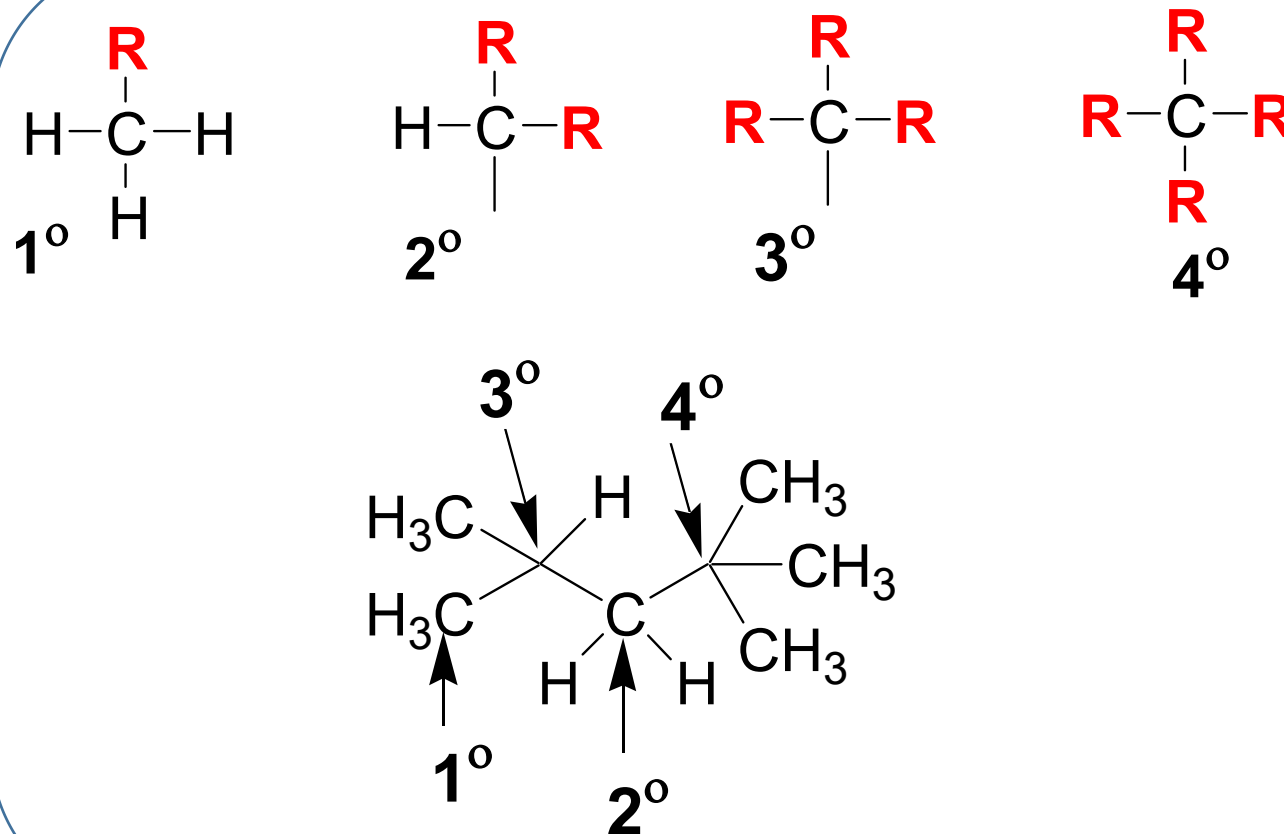
Radical Reaction Mechanism



Sauda Swaleh



Classification of Carbon Atoms



Assignment

Cyclopentane can be chlorinated to chlorocyclopentane using molecular chlorine in the presence of light.

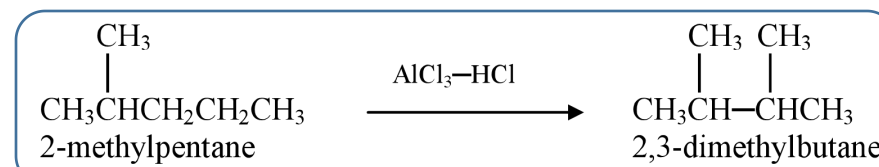
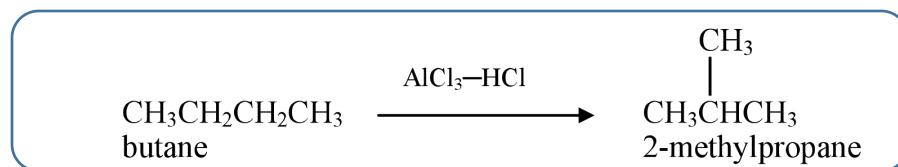
1. Write the initiation, propagation and termination steps for this free radical reaction.

2. Draw all the structures of the different monochlorinated products expected from the free radical chlorination of methylcyclobutane

Sauda
Swaleh

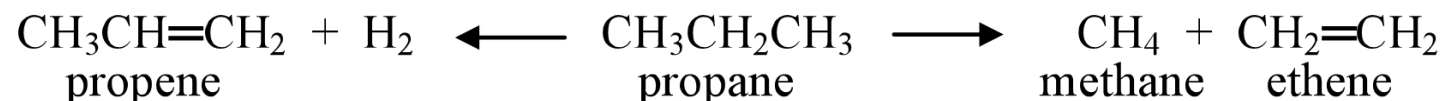
Isomerisation

- The molecular rearrangement of one compound into another compound or into more than one compound. Isomers with same MF but different structural formulae



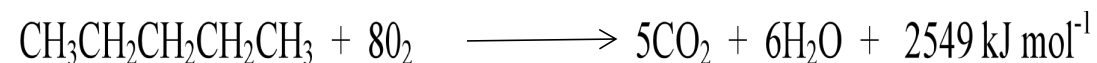
Pyrolysis

a process of decomposing an organic substance by heating it to high temperature in the absence of oxygen.



Combustion

Alkanes burn in excess of air or oxygen to give carbon dioxide and water



Task - ALKENES

a) Watch video links on

- i) <https://www.youtube.com/watch?v=RgbQya8HXBm> by JG the organic chemistry tutor
- ii) <https://www.youtube.com/watch?v=frtnEDTSzi8> by JG chemistry tutor
- b) Read through slides provided

Alkenes – Unsaturated Hydrocarbons

Alkenes

- Members of the alkene group have a double bond between two carbon atoms.
- One hydrogen atom has been removed from two adjacent carbon atoms, thereby allowing the two adjacent carbon atoms to form a double bond.
- General formula is C_nH_{2n}
- Begins with ethene (ethylene)
- C_2H_4

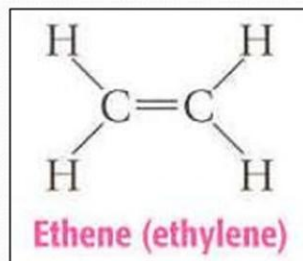
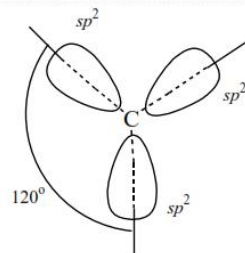


Figure 4.1 sp^2 hybridisation.

Natural Occurrence

- Plant products and in petroleum
- Play many important roles in biological systems – Hormones: Ethene - seed germination, flower maturation, and fruit ripening
- Pheromones released by insects
- Flavours and fragrances
- In industries – Alkenes are used as monomers to prepare polymers and solvents

Some Members of the Alkene Series

Name	Molecular Formula	Condensed Structural Formula
Ethene (ethylene)	C_2H_4	$CH_2=CH_2$
Propene	C_3H_6	$CH_3CH=CH_2$
1-Butene	C_4H_8	$CH_3CH_2CH=CH_2$
2-Butene	C_4H_8	$CH_3CH=CHCH_3$
1-Pentene	C_5H_{10}	$CH_3(CH_2)_2CH=CH_2$

Classification

Alkenes can be classified on the basis of the number of double bonds present in the molecule

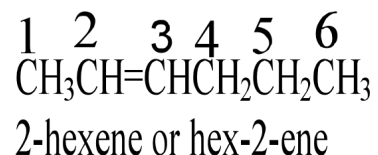
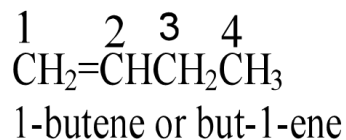
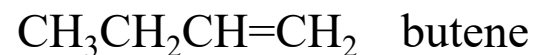
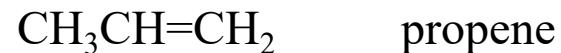
1. Monoenes- one double bond $C_nH_{2n} -C=C-$
2. Dienes- Two double bonds

Classes of Dienes (3)

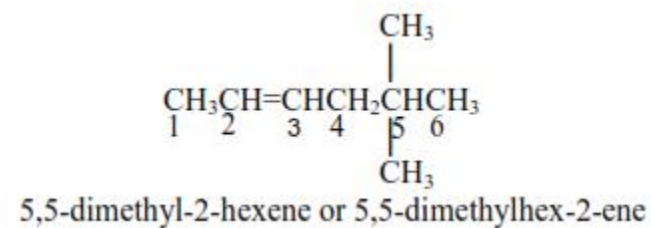
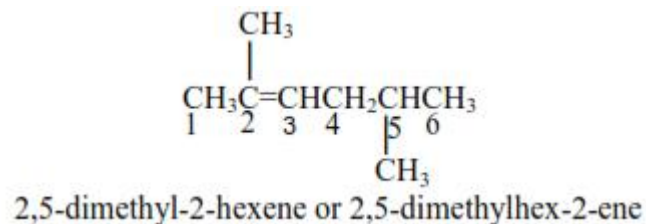
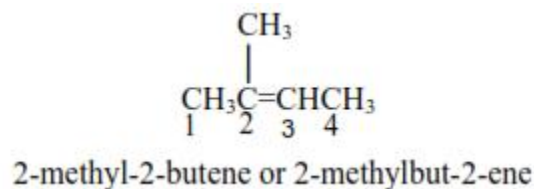
- a) Conjugated diene $-C=C-C=C-$ (Alternate single and double bonds)
- b)
 - i) Isolated diene $-C=C-C-C=C-$ (2 single bonds between double bond)
 - ii) Isolated diene $-C=C-C-C-C=C-$ (3 single bonds between double bond)
- c) Cumulated diene $-C=C=C-$ (allene)

Nomenclature

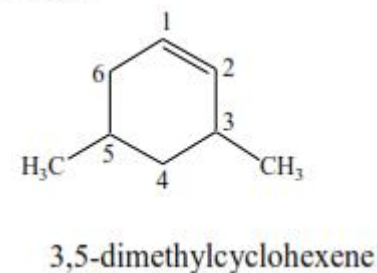
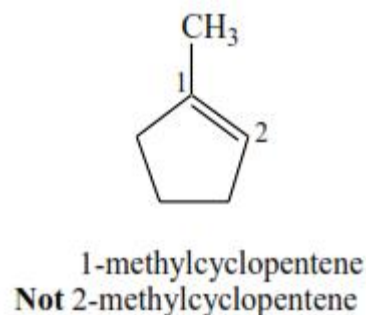
- “-ane” suffix for the corresponding alkane is changed to “-ene” for alkenes.
- A number preceding the name indicates the C atom on which the double bond starts.
 - *The carbons are numbered such that the double bond has the lowest number.*



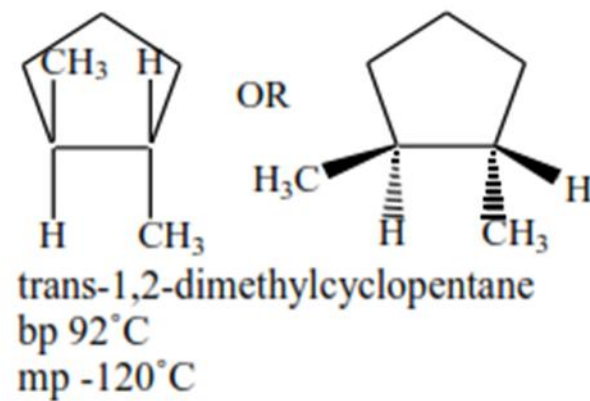
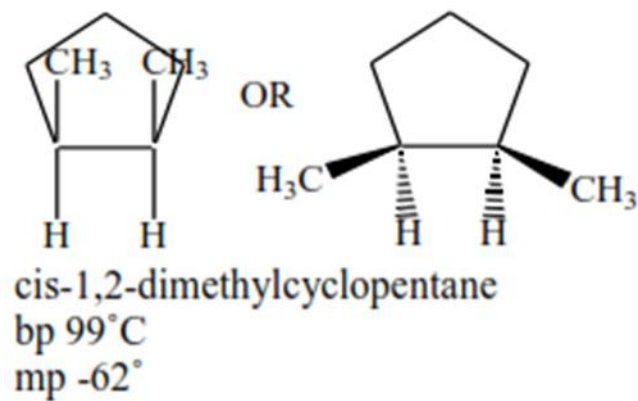
- The locations of substituent groups are indicated by the number of the carbon atoms to which they are attached.



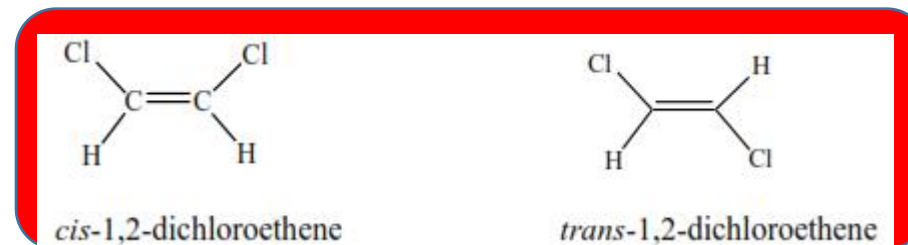
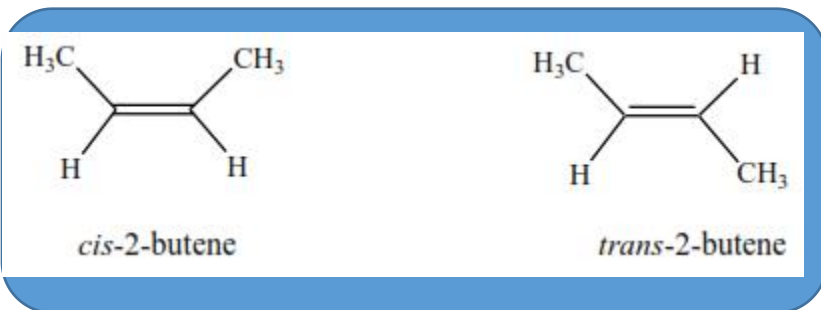
Substituted Cycloalkenes



Recap on Cis and trans Isomers

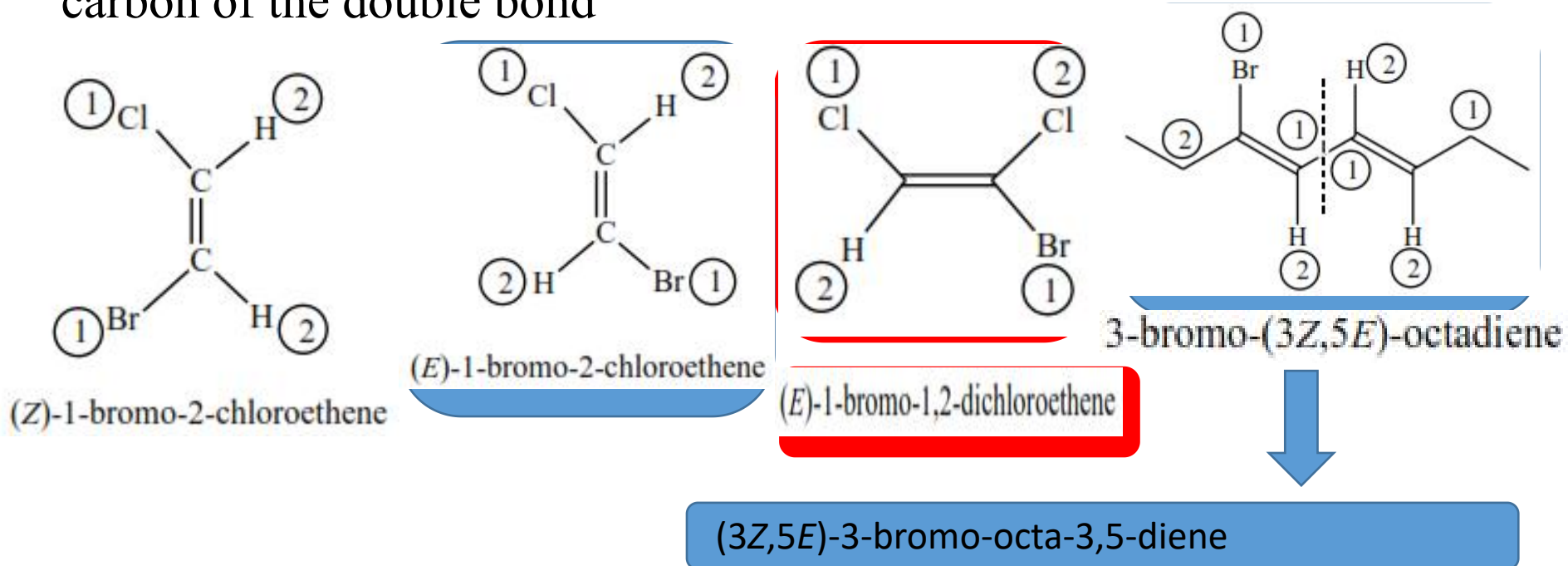


Geometrical isomers – Disubstituted Alkenes



For **tri- and tetrasubstituted** alkenes, the **Cahn-Ingold-Prelog** convention (sequence rule) is used.

- Assign priority based on atomic number of the atoms bonded on each carbon of the double bond



Physical properties:

- Carbon-carbon double bond changes the physicals properties of alkenes.
- At R.T. , alkenes exist in all three phases, solid, liquids, and gases.
- 1. Physical state:
- Ethene, Propene, and Butene exists as colorless gases.
- Members of the 5 or more carbons such as Pentene, Hexene, and Heptene are liquid

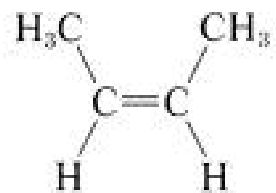
Note:

- Branching in an alkene also lowers the boiling point.
- Lower alkenes, from ethene to butene, are colourless gases,
- Higher ones, from C₅-C₁₅, are liquids and the rest are solids at room temperature.
- Alkenes are slightly more soluble in water than the corresponding alkanes because the π -electrons are attracted to the partially positive hydrogen of the water molecules.

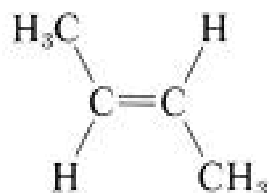
Recap

- Alkenes and their importance
- Classification of Alkenes- Monoenes, Dienes: Isolated, conjugated and Cumulative
- Nomenclature of Alkenes
- Di-substituted Alkenes (geometrical Isomers-Cis and trans Alkenes)
- Tri and tetra substituted Alkenes (*E* and *Z*) – assign priority about >C=C<

Recap – Geometrical Isomers

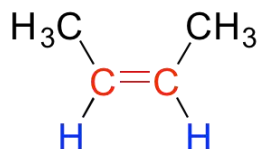


cis-2-butene



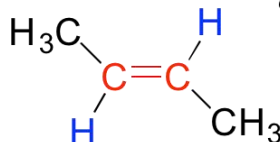
trans-2-butene

hydrogen atoms on same side

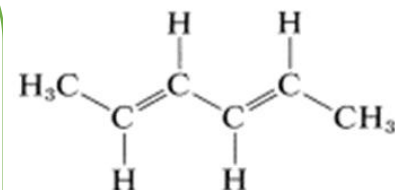


cis-alkene

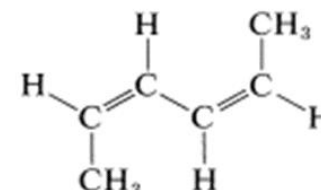
hydrogen atoms on opposite sides



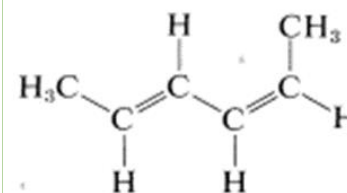
trans-alkene



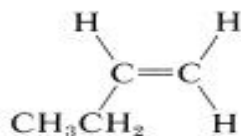
trans-trans-2,4-hexadiene



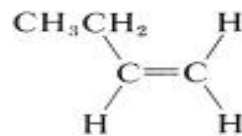
cis-cis-2,4-hexadiene



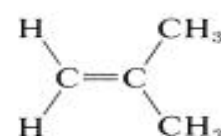
trans-cis- or cis-trans-2,4-hexadiene



identical to

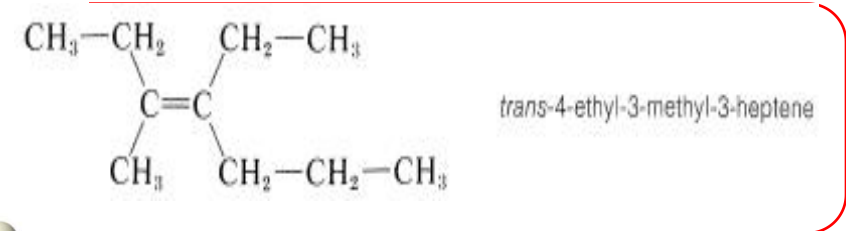
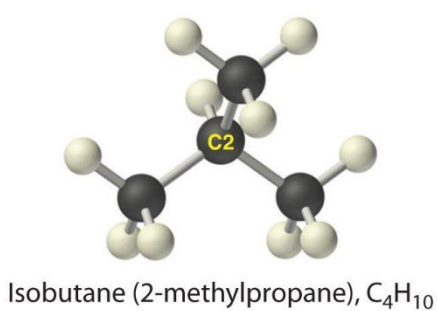
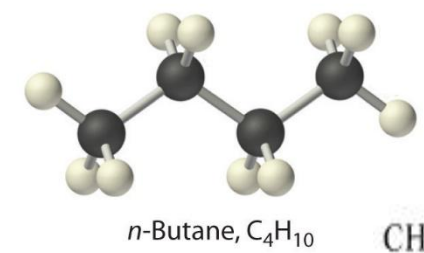
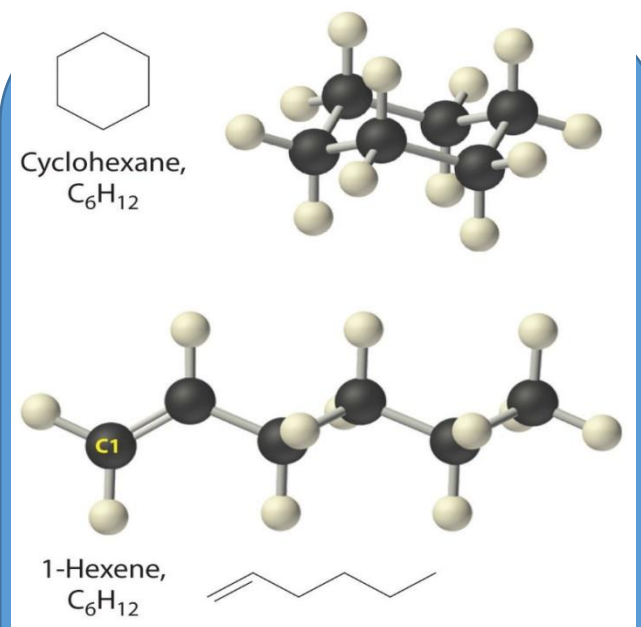
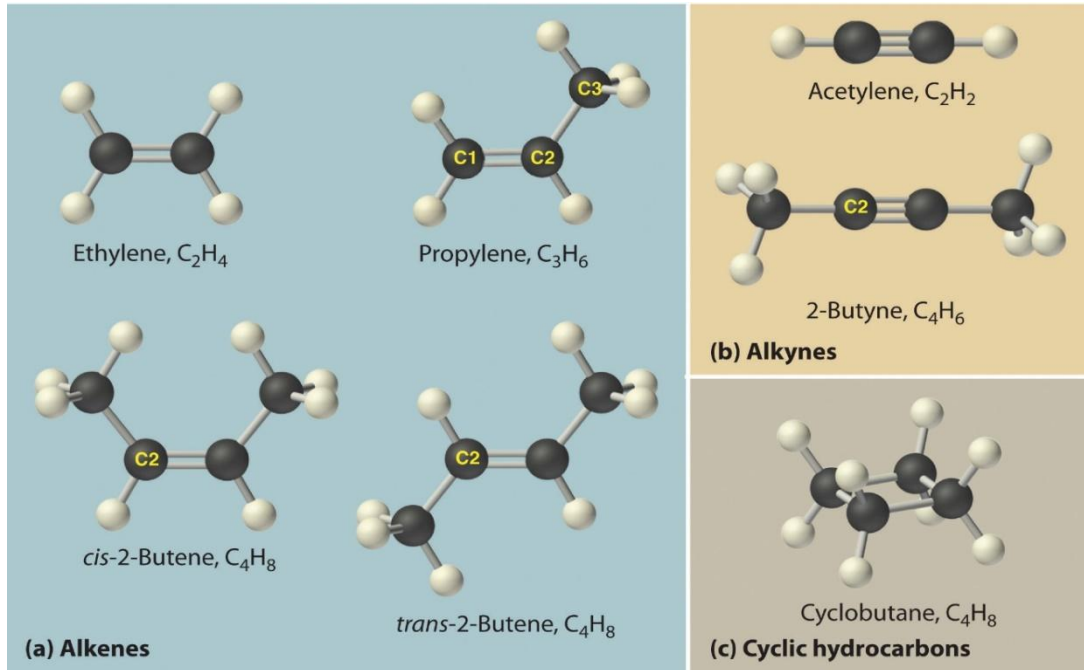


1-butene



2-methylpropene

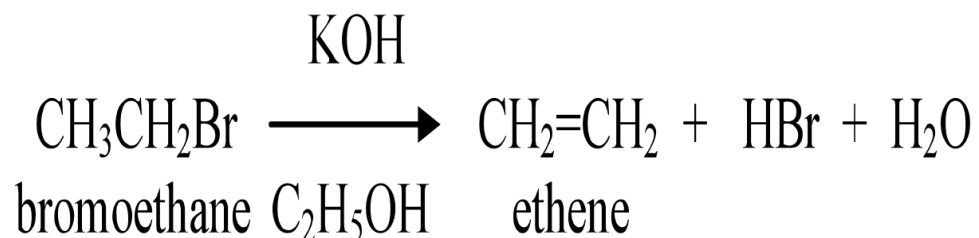
MF: C₄H₈



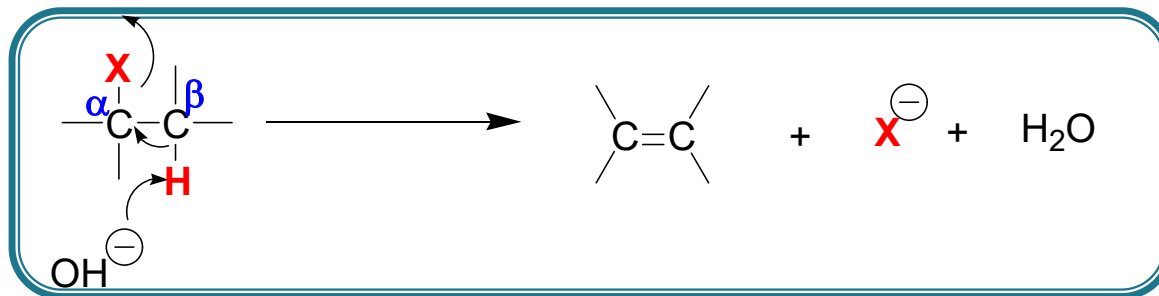
MF C6H12

Preparation of Alkenes

Dehydrohalogenation of Alkyl Halides



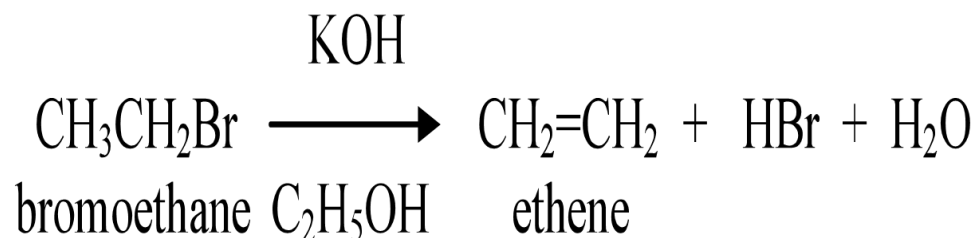
Ease of
dehydrohalogenation:
tert > sec > pr



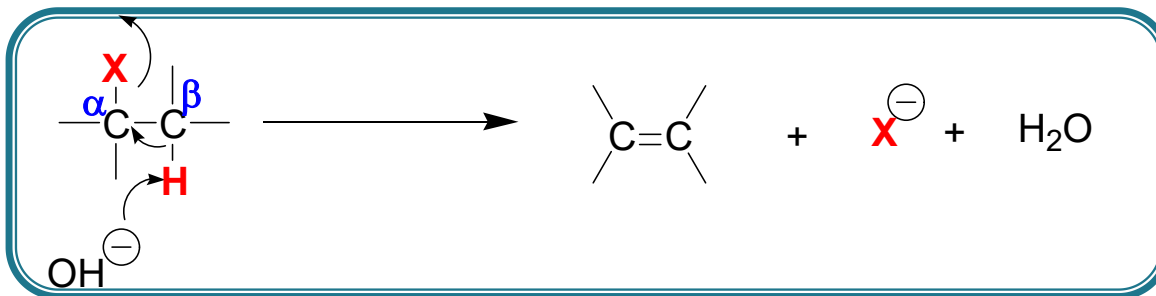
Note: Dehydrohalogenation follows **Saytzeff** or **Zaitsev's** rule = Hydrogen eliminated from Carbon with fewer number of Hydrogens to give highly substituted alkene as Major Product

Preparation of Alkenes

Dehydrohalogenation of Alkyl Halides

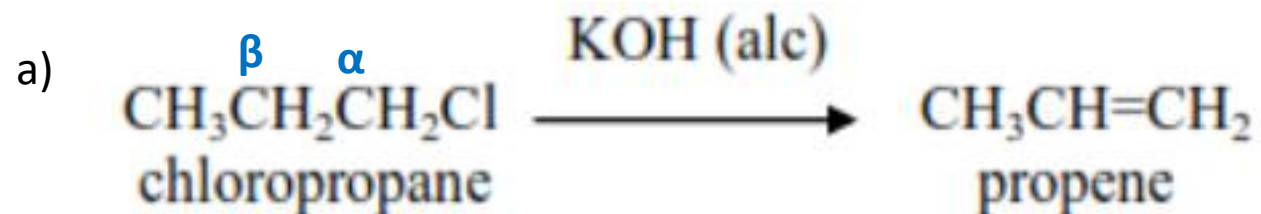


Ease of
dehydrohalogenation:
tert > sec > pr

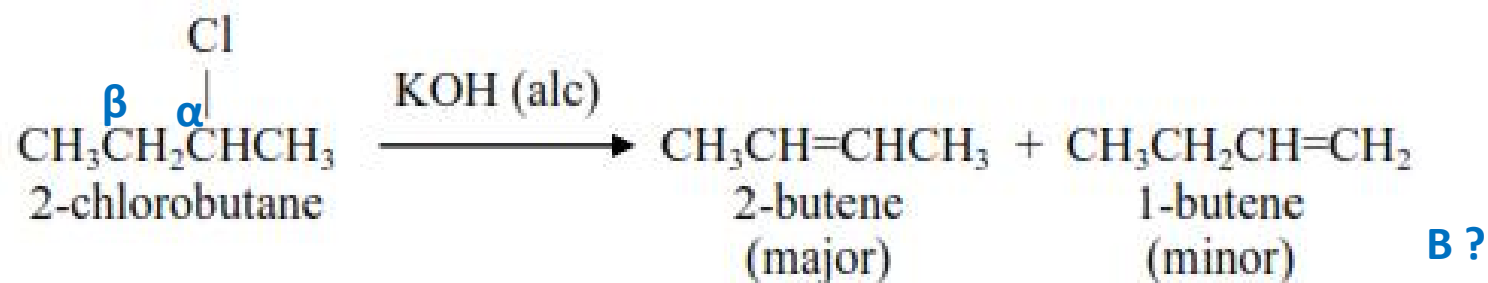


Note: Dehydrohalogenation follows **Saytzeff** or **Zaitsev's** rule = Hydrogen eliminated from Carbon with fewer number of Hydrogens to give highly substituted alkene as Major Product

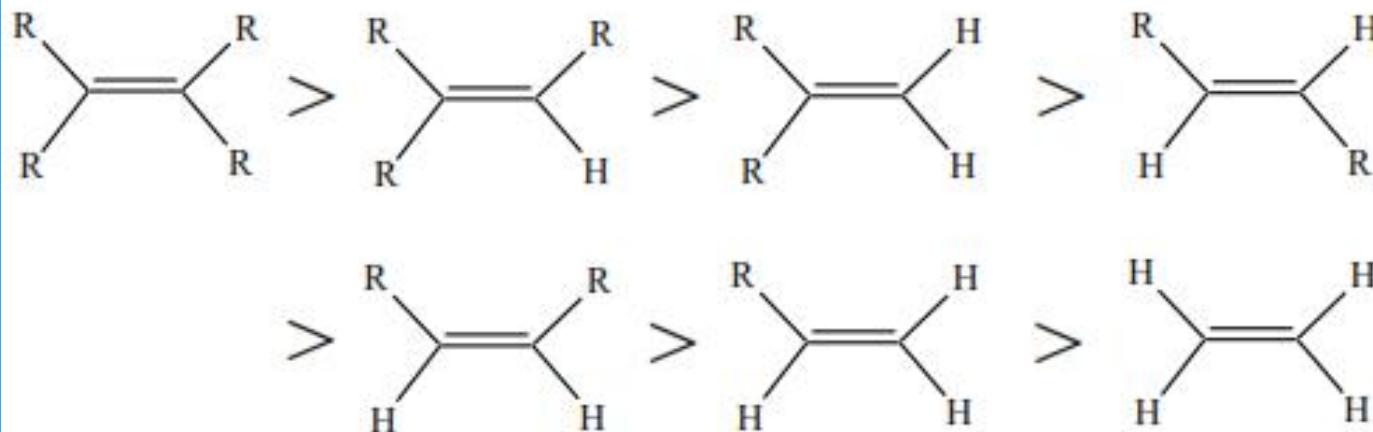
Halogen attached to terminal carbon



b) Halogen atom attached to non-terminal carbon atom



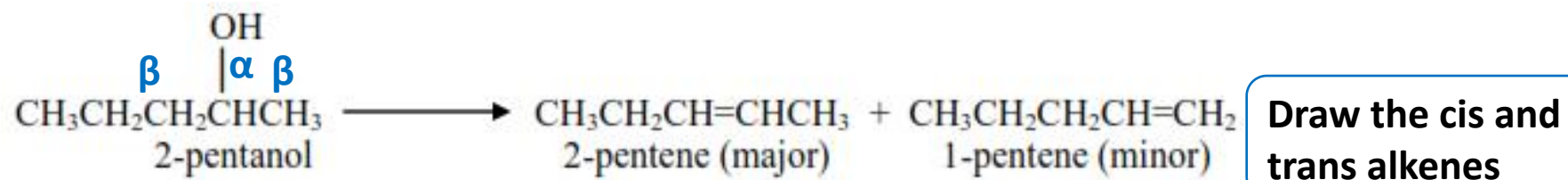
The stability of alkenes follows the following sequence:



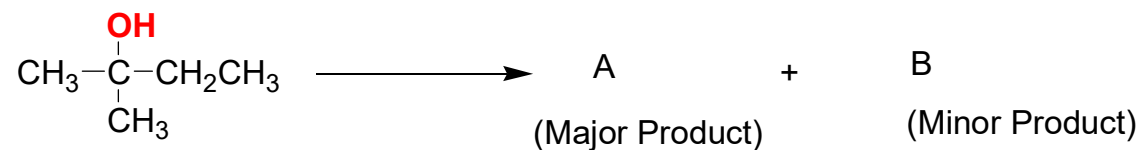
Dehydration of Alcohols – Removal of Water (H₂O) Molecule



II. Secondary alcohol (2°):

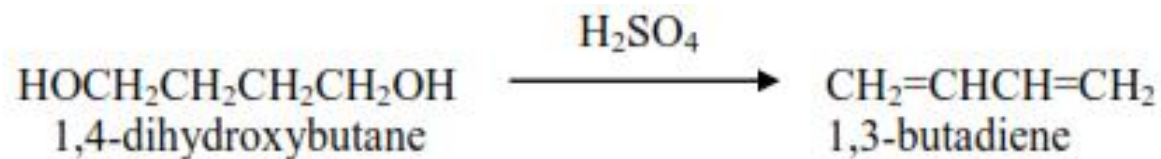


III. Tertiary alcohol (3°)



NB: Ease of dehydration of various alcohols follow the order: tert > sec > pr

Preparation of Dienes

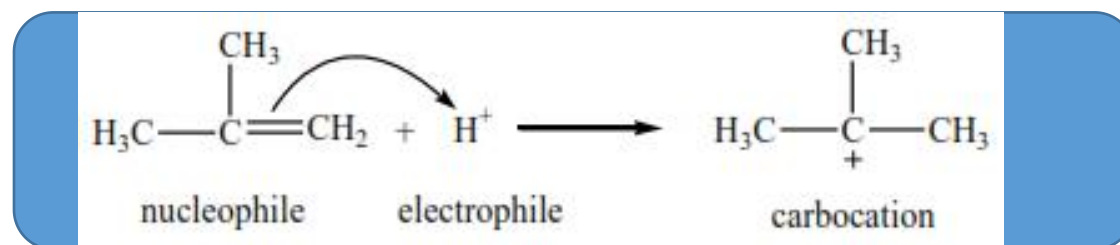


Reactions of alkenes

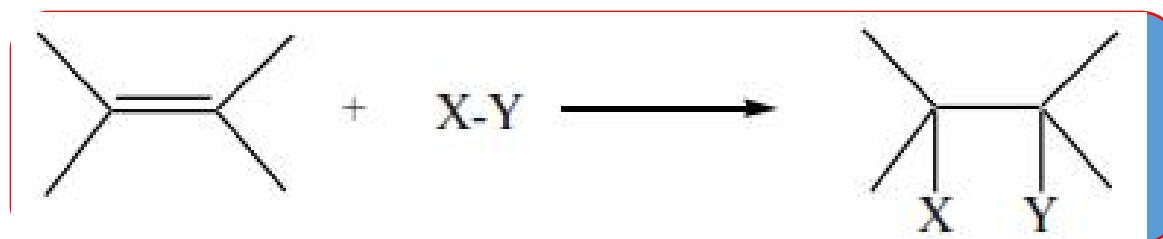
A nucleophile (Electron Rich Reagent): a nucleus loving reagent and has either a negative charge or a lone pair of electrons (**OH⁻**, **X⁻** or **H₂O**, **NH₃**) .

An electrophile: an electron loving reagent and has either a positive charge or an electron deficient system within it (**H⁺**) .

- When writing a **reaction mechanism** involving electrophiles and nucleophiles, a curly arrow is used to indicate the movement of a pair of electrons.
- Carbocation:** a species with a positive charge on a carbon atom



- Carbanion:** A species with a negative charge on a carbon atom
- Double bond** consists of a strong **σ** bond and a **π** weak bond



Electrophilic Addition
Reactions